

Identities, Multiple Angles and Exact Solution Sets

Combining Pythagorean identities, sum and difference formulas, double angles, and exact solution sets on a restricted interval

Directions.

- Every answer is exact: keep radicals and fractions, and do not round.
- For an equation, give *every* solution on the stated interval, in increasing order.
- The four tools — Pythagorean identities, sum and difference formulas, double-angle formulas, and factoring to solve — are mixed on purpose. Decide which one the problem needs before you start.

1. Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$ to a single trigonometric function.

Answer: _____

2. Use a sum or difference formula to find the exact value of $\cos 15^\circ$.

Answer: _____

3. An angle θ in the first quadrant has $\sin \theta = \frac{3}{5}$. Find the exact value of $\sin 2\theta$.

Answer: _____

4. Solve $2 \cos \theta - \sqrt{3} = 0$ for $0^\circ \leq \theta < 360^\circ$.

Answer: _____ degrees

5. Use a difference formula to find the exact value of $\tan 15^\circ$. Rationalise the denominator so that no radical remains below the line.

Answer: _____

6. Solve $\sin 2\theta = \sin \theta$ for $0^\circ \leq \theta < 360^\circ$. Give all four solutions.

Answer: _____ degrees

7. Simplify $\frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta}$ to a single fraction with one trigonometric function in it.

Answer: _____

8. Solve $2 \cos^2 \theta + \cos \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Answer: _____ degrees

9. Solve $\sin 2\theta = \cos \theta$ for $0^\circ \leq \theta < 360^\circ$. Give all four solutions.

Answer: _____ degrees

10. Find the exact value of $\cos 105^\circ$ by writing 105° as a sum of two special angles.

Answer: _____

11. Consider the identity $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = \frac{2}{\sin \theta}$.

- (a) Combine the left-hand side over a common denominator and simplify it to a single term.

Answer: _____

- (b) Prove the identity, working on the left-hand side only and naming the identity used at each step.

Answer: _____

12. Consider $\cos 2\theta + 3 \sin \theta = 2$ on $0^\circ \leq \theta < 360^\circ$.

- (a) Use a double-angle formula to rewrite the equation, then factor the left-hand side of the resulting quadratic.

Answer: _____

- (b) Give the exact solution set on the interval.

Answer: _____ degrees

- (c) Explain why this equation has exactly three solutions on the interval rather than four.

Answer: _____